X-Ray Diffraction: Lattice Spacings of LiF and KBr and Planck's Constant from the Bremsstrahlung Cut-off

Parth Bhargava
PC2193, Experimental Physics I
National University of Singapore
Semester 1, AY2025/26

Copper K_α and K_β radiation diffracted by a single crystal measures the spacing of its lattice planes through Bragg's law, and the short-wavelength limit of the continuous spectrum, read off the same crystal, measures Planck's constant. Repeated θ–2θ scans of the (200) reflections of LiF and of a crystal supplied as KBr were fitted with line-shape models, and each crystal's spacing and goniometer zero offset were obtained together from its four reflections. For KBr the offset model described the peaks within their uncertainties and gave $d = (3.302 \pm 0.001)$ Å, a lattice constant of (6.604 ± 0.002) Å consistent with the tabulated 6.60 Å for KBr. For LiF it did not ($\chi^2_\nu = 20$): the apparent spacing fell from 2.068 to 2.034 Å with increasing angle, and the offset-corrected $d = (2.024 \pm 0.003)$ Å remains 0.7% above the tabulated 2.01 Å. At 25 kV the bremsstrahlung edge, located by a background-plus-linear-rise fit, gave $\lambda_{\min} = (0.475 \pm 0.011)$ Å and $h = (6.35 \pm 0.15) \times 10^{-34}$ J·s, 1.9 standard deviations below the defined value. At 30 and 35 kV the edge lay at or below the start of the scan, so only an upper limit (30 kV) or no value (35 kV) could be obtained. The angle calibration is the main limitation: without the offset correction, h would be 5% lower.

1. INTRODUCTION

The atoms of a crystal sit on lattice planes a few ångström apart, comparable with the wavelength of X-rays. A crystal therefore acts as a diffraction grating for X-rays: a monochromatic beam is reflected strongly only at angles fixed by the plane spacing, which makes X-ray diffraction the standard way to measure crystal structure [1]. The same crystal, used in reverse with a known spacing, disperses an X-ray spectrum by wavelength. The continuous spectrum of an X-ray tube ends abruptly at a shortest wavelength fixed by the tube voltage, the Duane–Hunt limit [2], which relates the photon energy to the electron energy and so measures Planck's constant.

This experiment used a copper-anode tube and a θ–2θ goniometer to (i) measure the (200) spacing of LiF from the angles of its K_α and K_β reflections in two orders, (ii) measure the spacing of a second crystal, supplied as KBr, and compare it with tabulated alkali halides, and (iii) locate the short-wavelength limit of the bremsstrahlung, diffracted by LiF, at three tube voltages to determine h [3].

2. THEORY

2.1 Bragg Reflection

X-rays are scattered elastically by the electrons of every atom. For a set of parallel planes of spacing d, the waves reflected from successive planes differ in path by $2d \sin\theta$, where θ is the glancing angle between the beam and the planes (Fig. 1). They interfere constructively when

$$n\lambda = 2d \sin\theta, \tag{1}$$

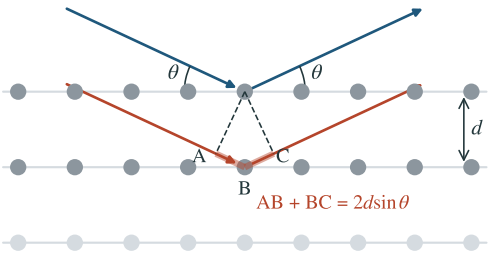


FIG. 1. Bragg reflection from lattice planes a distance d apart. Rays meet the planes at the glancing angle θ; the ray reflected from the lower plane travels the extra path AB + BC = $2d \sin\theta$, and the reflections add in phase when this is a whole number of wavelengths.

with n the order of reflection [1,3]. Each reflection with known n and λ gives an apparent spacing $d_i = n\lambda/(2 \sin\theta_i)$. An error σ_θ in the Bragg angle propagates as

$$\sigma_d/d = \cot\theta\, \sigma_\theta, \tag{2}$$

with σ_θ in radians, so low-angle reflections are the most sensitive to angle errors.

LiF and KBr both have the rock-salt structure, in which the (200) planes of a cubic lattice of constant a are $d = a/2$ apart. The tabulated lattice constants are 4.02 Å for LiF and 6.60 Å for KBr [4]; the manual quotes $d = 2.008$ Å for LiF [3].

2.2 Copper Emission Lines

Electrons accelerated through a potential V strike the copper anode and produce two kinds of radiation [1,3]. Ionisation of the K shell followed by the fall of an L or M electron gives the characteristic lines: $K_{\alpha1}$ (1.54056 Å) and $K_{\alpha2}$ (1.54439 Å), and K_β (1.39222 Å) [5]. $K_{\alpha1}$ is about twice as intense as $K_{\alpha2}$

[1]. The doublet is unresolved at low angles but separates by $\Delta(2\theta) = 2\tan\theta\,\Delta\lambda/\lambda$, which is 0.33° for the second-order LiF reflection, so it is fitted as two lines (Sec. 3.2).

2.3 Bremsstrahlung Cut-off

Deceleration of the electrons in the anode also produces a continuous spectrum. A photon cannot carry more than the electron's kinetic energy eV, so the spectrum ends at

$$\lambda_{\min} = \frac{hc}{eV}, \quad \text{i.e.} \quad h = \frac{eV\lambda_{\min}}{c}, \tag{3}$$

with h in J·s for V in volts and $\lambda_{\min}$ in metres [2,3]. Kramers' thick-target law gives an intensity per unit frequency proportional to $\nu_0 - \nu$, where $h\nu_0 = eV$ [6]. Per unit wavelength this is proportional to $(\lambda - \lambda_{\min})/(\lambda^2\lambda_{\min})$, which rises linearly from zero just above the cut-off. The crystal maps λ to the detector angle through Eq. (1) with $n = 1$, and near the cut-off ($2\theta \approx 10$ –15°) $\mathrm{d}\lambda/\,\mathrm{d}(2\theta) = d\cos\theta$ changes by less than 1%. The count rate is therefore modelled as a constant background b, from scattered and higher-order radiation, below an edge at $2\theta_c$ and a linear rise above it:

$$R(2\theta) = b + k\max(0, 2\theta - 2\theta_c). \tag{4}$$

The edge gives $\lambda_{\min} = 2d\sin\theta_c$ and, through Eq. (3), h. This replaces extrapolating the rise to zero count rate, which places the edge too low when $b > 0$.

2.4 Goniometer Zero Offset

If the angle scale reads δ too high, every measured detector angle is $2\theta_i = 2\arcsin(n_i\lambda_i/2d) + \delta$. Then by Eq. (2) the apparent spacings follow $d_i \approx d(1 - \cot\theta_i\,\delta/2)$: they lie on a line against $\cot\theta_i$ that meets the true d at $\cot\theta = 0$. Fitting d and δ together to four reflections of different angle separates the two, and the quality of the fit tests whether a constant offset is the only angle error.

3. METHODS

3.1 Apparatus and Scans

Measurements used an integrated PHYWE 09058.99 X-ray unit with a copper-anode tube operated at 0.05 mA, a goniometer in 2:1 (θ–2θ) coupling and a Geiger–Müller counter, under computer control [3] (Fig. 2). The software scanned the crystal angle and recorded the count rate, rounded to an integer number per second, against the detector angle 2θ. Table I lists the scans. Survey scans located the reflections; narrow scans of each reflection, repeated five or six times, gave the peak positions; and the bremsstrahlung was recorded at 25, 30 and 35 kV in one multi-voltage scan, repeated five times. All scans were taken with a 35 kV tube voltage except the bremsstrahlung scans.

The exported angles are truncated rather than rounded to one decimal (35.6° appears as 35.5°, 80.2° as 80.1°). Every scan was checked to lie on a uniform grid within this truncation, and the grid values were used. One repeat of the second-order LiF scan was recorded against the crystal angle θ and was converted to 2θ.



FIG. 2. The PHYWE 09058.99 X-ray unit. The tube sits behind the window on the left; the goniometer in the central chamber carries the crystal at its centre and the Geiger–Müller counter on its arm; the unit is controlled from the computer on the right. Photograph: PHYWE [7].

TABLE I. Scans. Step and range are in detector angle; the crystal turned through half of each. Time is the counting time per point.

Scan	2θ range (°)	Step, time	Repeats
Survey, LiF and KBr	10–110	1°, 5 s	1
LiF, first order	35–55	0.2°, 1 s	6
LiF, second order	80–105	0.2°, 1 s	5
KBr, first order	20–30	0.2°, 1 s	5
KBr, second order	40–60	0.2°, 1 s	5
Bremsstrahlung, LiF	10–40	0.2°, 1 s	5

3.2 Analysis

Each narrow scan was fitted separately by weighted least squares. A K_β peak was modelled as a Gaussian on a linear background. A K_α peak was modelled as a $K_{\alpha1}$–$K_{\alpha2}$ doublet of equal widths, with the intensity ratio fixed at 2:1 and the $K_{\alpha2}$ position tied to $K_{\alpha1}$ through Eq. (1); its fitted position is that of $K_{\alpha1}$. With 1 s counting the rates are counts, so each point was weighted by $\sqrt{\max(R, 1)}$. The Poisson assumption was checked on the bremsstrahlung repeats, where the variance across scans was 0.99–1.07 times the mean. The peak position is the mean of the fitted centres over the repeats, and its uncertainty is their standard error, which includes scan-to-scan positioning scatter as well as counting noise. Each position gave d_i by Eq. (1), with uncertainty from Eq. (2). For each crystal, d and δ were then fitted to the four positions, weighted by their standard errors, and the covariance was scaled by $\sqrt{\chi^2_\nu}$ where $\chi^2_\nu > 1$. Moving every fit window edge by ±0.6° tested the sensitivity to the windows.

The manual prescribes the intensity-weighted mean angle of each peak (centroid) [3]. Without background subtraction, the centroid is pulled towards the centre of the window and averages the doublet. It was computed only for comparison.

For the bremsstrahlung, the five repeats were averaged and Eq. (4) was fitted from the start of the scan (10°) to 17° at 25 kV and 15° at 30 kV, covering the linear rise but not the turnover beyond it. Because χ^2 is not smooth in $2\theta_c$, the edge was located by scanning $2\theta_c$ in 0.005° steps and solving for b

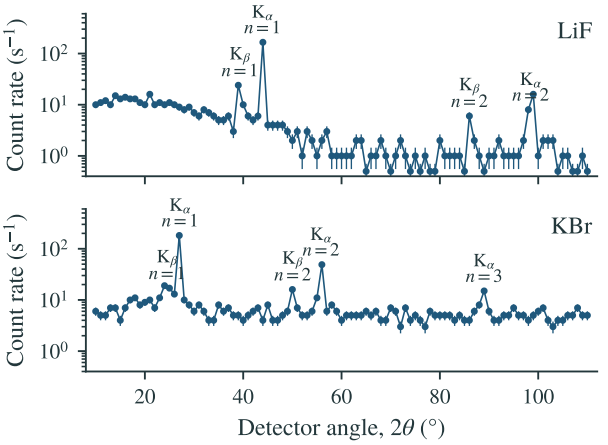


FIG. 3. Survey scans of LiF and KBr at 35 kV (1° steps, 5 s per point; logarithmic rate axis, error bars from counting statistics). Labels give the line and order of each (200)-family reflection, placed at the fitted positions of Table II; the KBr third-order K_α label is at its predicted position.

and k by linear least squares at each step. The 1σ interval is where χ^2 rises by one above its minimum (by χ^2_ν if $\chi^2_\nu > 1$). The edge was converted with the LiF values of d and δ; a Monte Carlo over the edge interval and the d–δ covariance (20 000 draws) gave the uncertainties of $\lambda_{\min}$ and h. The upper window limit was varied over $\pm 1°$ as a sensitivity test. The tube-voltage calibration could not be checked and is not included.

4. RESULTS

4.1 Diffraction Patterns

Both crystals showed the same pattern of four strong reflections: K_β and K_α in first and second order (Fig. 3). In each order K_β lay at the lower angle, as its shorter wavelength requires, and first-order reflections were the stronger. The KBr reflections lay at smaller angles than those of LiF, so its spacing is larger. KBr also showed a third-order K_α reflection: the survey maximum at 89° agrees, within its 1° step, with the 89.1° predicted from the fitted spacing below. For LiF the third order would need $\sin\theta > 1$. The LiF background below 35° is the bremsstrahlung analysed in Sec. 4.3.

4.2 Peak Positions and Lattice Spacings

The line-shape models described the individual scans with χ^2_ν between 0.4 and 2.6, except two first-order KBr scans at 6.0 and 8.3 (Fig. 4). The fitted centres of repeated scans scattered more than their fit errors for some reflections. The largest scatter was for the second-order LiF K_α reflection, whose centre moved steadily from 98.62° to 98.32° over its five repeats; this scatter sets its standard error. The weakest reflection, second-order KBr K_β, stood only about ten counts per second above the background. Table II lists the positions and apparent spacings.

The apparent spacings were not constant (Table II). For LiF they fell from 2.068 to 2.034 Å as the angle increased; their mean, 2.050 Å with a standard deviation of 0.016 Å, is far less precise than any single value. For KBr they rose from 3.274 to 3.291 Å. A trend with angle is the signature of an angle error

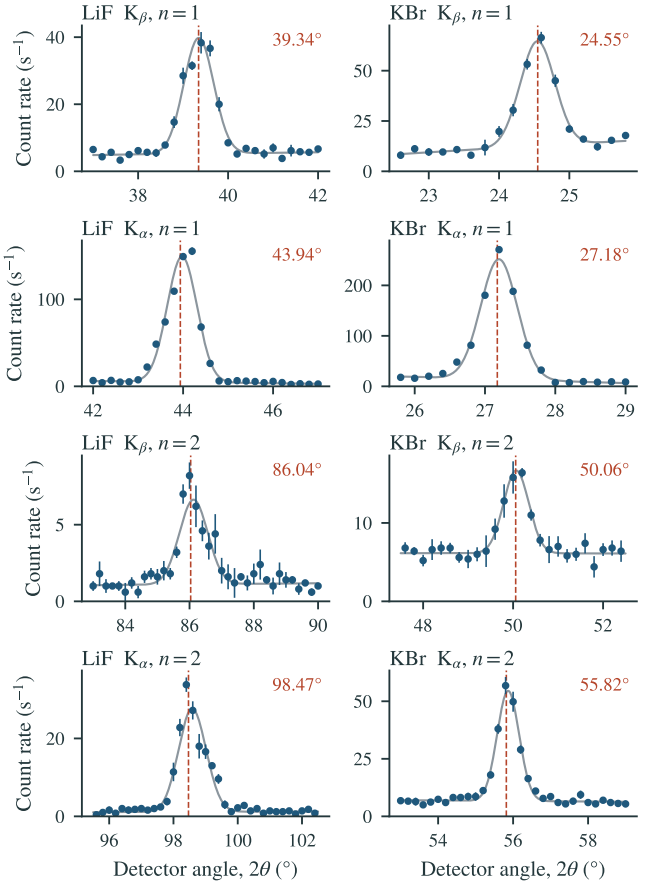


FIG. 4. Narrow scans of the four reflections of LiF (left) and KBr (right). Points are the mean count rate over the repeated scans with its standard error; grey curves are the line-shape model (Gaussian for K_β, $K_{\alpha1}$–$K_{\alpha2}$ doublet for K_α) fitted to the mean profile; dashed lines and labels give the mean of the fitted centres of the individual scans ($K_{\alpha1}$ for the doublets). The $K_{\alpha2}$ shoulder is visible on the high-angle side of the second-order LiF reflection.

(Eq. (2)), and its opposite sign in the two crystals shows that the error was not a fixed property of the angle scale.

For KBr the zero-offset model described all four positions ($\chi^2_\nu = 0.37$; Fig. 5). It gave $d = (3.302 \pm 0.001)$ Å and $\delta = (+0.20 \pm 0.01)°$, and the window test moved d by 0.001 Å.

TABLE II. Peak positions (mean and standard error over six LiF first-order scans and five scans otherwise) and apparent spacings from Eq. (1) without offset correction. The last column applies the manual's centroid method to the same scans, using the intensity-weighted mean K_α wavelength.

Reflection	2θ (°)	d_i (Å)	**Centroid** d_i (Å)
LiF			
K_β, $n=1$	39.34 ± 0.02	2.0678 ± 0.0009	2.062
$K_{\alpha1}$, $n=1$	43.938 ± 0.005	2.0590 ± 0.0002	2.057
K_β, $n=2$	86.04 ± 0.03	2.0407 ± 0.0005	2.034
$K_{\alpha1}$, $n=2$	98.47 ± 0.05	2.0341 ± 0.0008	2.032
KBr			
K_β, $n=1$	24.55 ± 0.02	3.274 ± 0.002	3.288
$K_{\alpha1}$, $n=1$	27.175 ± 0.005	3.2787 ± 0.0005	3.282
K_β, $n=2$	50.06 ± 0.08	3.291 ± 0.005	3.293
$K_{\alpha1}$, $n=2$	55.82 ± 0.01	3.2913 ± 0.0005	3.289

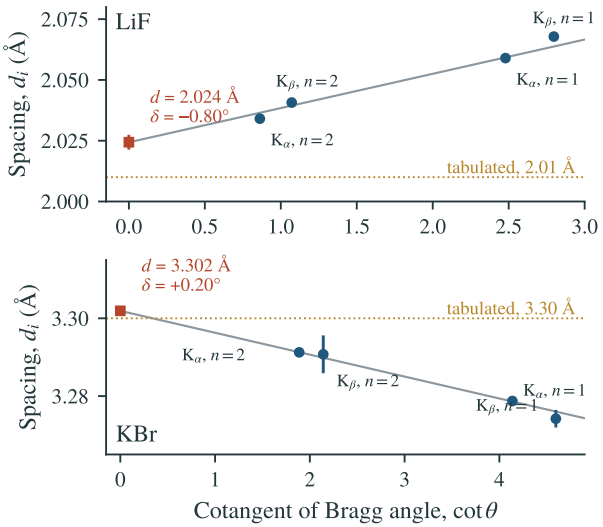


FIG. 5. Apparent spacing against cot θ for LiF (top) and KBr (bottom), with error bars from Table II. Grey curves are the joint fit of d and the zero offset δ; the red square is the fitted d at cot $\theta = 0$, and the dotted line the tabulated spacing [4].

The lattice constant $a = 2d = (6.604 \pm 0.002)$ Å agrees with the tabulated 6.60 Å for KBr [4].

For LiF the same model failed: $\chi^2_\nu = 19.6$. The residuals of the four positions were $-0.08°$, $+0.01°$, $-0.07°$ and $+0.16°$ in the order of Table II, up to three times the scatter of repeated scans. With its covariance scaled by $\sqrt{\chi^2_\nu} = 4.4$, the fit gave $d = (2.024 \pm 0.003)$ Å and $\delta = (-0.80 \pm 0.08)°$, with the two strongly correlated (correlation coefficient 0.97). The window test moved d by less than 0.001 Å. This d is 0.014 Å (0.7%) above the tabulated $a/2 = 2.01$ Å [4] and 0.016 Å above the manual's 2.008 Å [3]. The uncorrected mean of 2.050 Å was 2% above.

4.3 Bremsstrahlung Cut-off and Planck's Constant

All three spectra rose from a low background to a broad maximum and fell towards the K_β reflection (Fig. 6 a). The rise began at smaller angles for higher voltages, as Eq. (3) requires.

At 25 kV the background of 1.2 s^{-1} below the edge was sampled over 2.5°, and the fit ($\chi^2_\nu = 0.73$) located the edge at $2\theta_c = (12.69 \pm 0.31)°$ (Fig. 6 b). Varying the upper window limit from 16° to 18° moved it by less than 0.1°. With the LiF calibration this gives $\lambda_{\min} = (0.475 \pm 0.011)$ Å and

$$h = (6.35 \pm 0.15) \times 10^{-34}\ \text{J·s},$$

which is 4.2%, or 1.9 standard deviations, below the defined value 6.626×10^{-34} J·s [8]. The defined value of h predicts the edge at 13.27°.

At 30 kV the predicted edge, 10.92°, lies only 0.9° above the start of the scan, so the background is sampled by a few points. The best fit placed the edge at 10.2°, but χ^2 stayed within one unit of its minimum for every edge position below the scan start. The data give only the upper limit $2\theta_c < 10.9°$ (1σ), which corresponds to $h < 6.61 \times 10^{-34}$ J·s and is consistent with the defined value. At 35 kV the predicted edge, 9.24°, lies below the scan start, and the mean rate over the first five points, 10.2 s^{-1}, was well above the 25 kV background. No edge was in range.

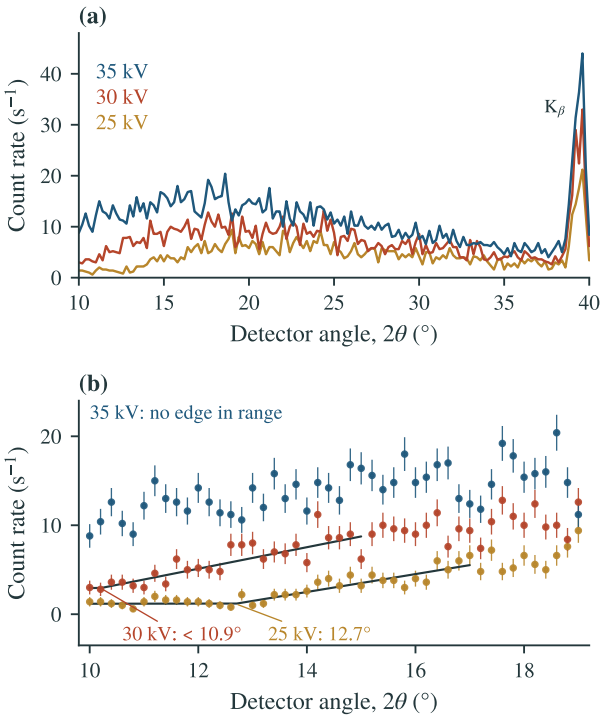


FIG. 6. Bremsstrahlung diffracted by LiF at 25, 30 and 35 kV, each the mean of five scans. (a) The continuous spectrum, ending at the first-order K_β reflection near 39°. (b) The low-angle region, with error bars from counting statistics. Black lines are fits of Eq. (4) (constant background, then linear rise) over the fit windows; labels give the edge $2\theta_c$ (for 30 kV, the 1σ upper limit). At 35 kV the rate was already well above background at the start of the scan.

5. DISCUSSION

The KBr result shows that the method reaches a precision of 0.03% in d when a constant offset accounts for the angle errors. The LiF reflections show a second, angle-dependent error. The residuals do not follow a smooth function of angle: they were negative for both K_β reflections and positive for both K_α reflections, and they exceed the scatter of repeated scans, so they are not counting noise. The steady drift of the second-order K_α centre by 0.3° across successive scans points to mechanical positioning of the goniometer, as does the larger scatter of the second-order LiF scans generally. A displacement of the crystal surface from the goniometer axis would also produce an angle-dependent shift. Four reflections cannot separate such terms from d and δ, so the cause was not identified. The consequence is that the LiF spacing is uncertain by about its 0.7% discrepancy from the tabulated value, not by the 0.15% of the scaled fit. The offset changed from $-0.80°$ to $+0.20°$ between the two crystals, so it depends on how each crystal was mounted and must be fitted for every mounting.

The KBr spacing matches KBr, which has the rock-salt structure and $a = 6.60$ Å. The isostructural RbCl ($a = 6.58$ Å [4]) differs by 0.3%, less than the 0.7% systematic discrepancy seen for LiF. The spacing alone therefore does not exclude it,

and the identification rests on the spacing together with the sample's labelling.

The value of h depends on the angle calibration more than on counting statistics. Relative to the uncorrected calibration, the fitted d and δ together raised $\lambda_{\min}$ by 3.4%: with the uncorrected mean spacing and no offset, the same edge gives $h = 6.05 \times 10^{-34}$ J·s, 8.6% low. The offset was fitted at 39–99° and extrapolated to 13°, where the LiF residuals show it may be wrong by up to about 0.1°. An error of 0.1° in $2\theta_c$ changes h by 0.8%, so this extrapolation, together with the unchecked tube-voltage calibration, may account for the 1.9σ shortfall. The measurement also depended on the scan range. The manual's starting crystal angle of 5° places the 35 kV edge below the scan and the 30 kV edge within a degree of its start, leaving one usable voltage. No test of $\lambda_{\min} \propto 1/V$ was therefore possible.

6. CONCLUSION

Bragg reflection of copper K radiation gave the (200) spacing of KBr as $d = (3.302 \pm 0.001)$ Å, a lattice constant of (6.604 ± 0.002) Å consistent with KBr, after a fitted goniometer offset of $(+0.20 \pm 0.01)°$. For LiF the offset model left residuals of up to 0.16°, and the offset-corrected $d = (2.024 \pm 0.003)$ Å remained 0.7% above the tabulated 2.01 Å. The bremsstrahlung edge at 25 kV gave $h = (6.35 \pm 0.15) \times 10^{-34}$ J·s, 1.9 standard deviations below the defined value; at 30 kV only an upper limit was obtained, and at 35 kV the edge lay outside the scan.

Three changes address these limitations. Starting the bremsstrahlung scan at a crystal angle of about 3.5° ($2\theta =$ 7°) would sample the background below the edges at all three voltages, giving three independent values of h and a test of $\lambda_{\min} \propto 1/V$. Measuring the direct beam at $2\theta = 0$, or each reflection on both sides of it if the goniometer permits, would measure the zero offset directly instead of extrapolating it from four reflections. Interleaving repeated scans of different reflections, rather than repeating one reflection consecutively, would turn the slow positioning drift seen in the second-order LiF scans into random scatter that the standard errors capture.

7. REFERENCES

[1] B. D. Cullity and S. R. Stock, *Elements of X-Ray Diffraction*, 3rd ed. (Prentice Hall, 2001).

[2] W. Duane and F. L. Hunt, On X-ray wave-lengths, Phys. Rev. **6**, 166 (1915).

[3] Department of Physics, National University of Singapore, X-Ray Diffraction of Crystals, (2025).

[4] N. W. Ashcroft and N. D. Mermin, *Solid State Physics* (Holt, Rinehart, Winston, New York, 1976).

[5] J. A. Bearden, X-ray wavelengths, Rev. Mod. Phys. **39**, 78 (1967).

[6] H. A. Kramers, On the theory of X-ray absorption and of the continuous X-ray spectrum, Phil. Mag. **46**, 836 (1923).

[7] PHYWE Systeme, X-ray unit, 35 kV (09058.99), (n.d.).

[8] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, CODATA recommended values of the fundamental physical constants: 2018, Rev. Mod. Phys. **93**, 25010 (2021).